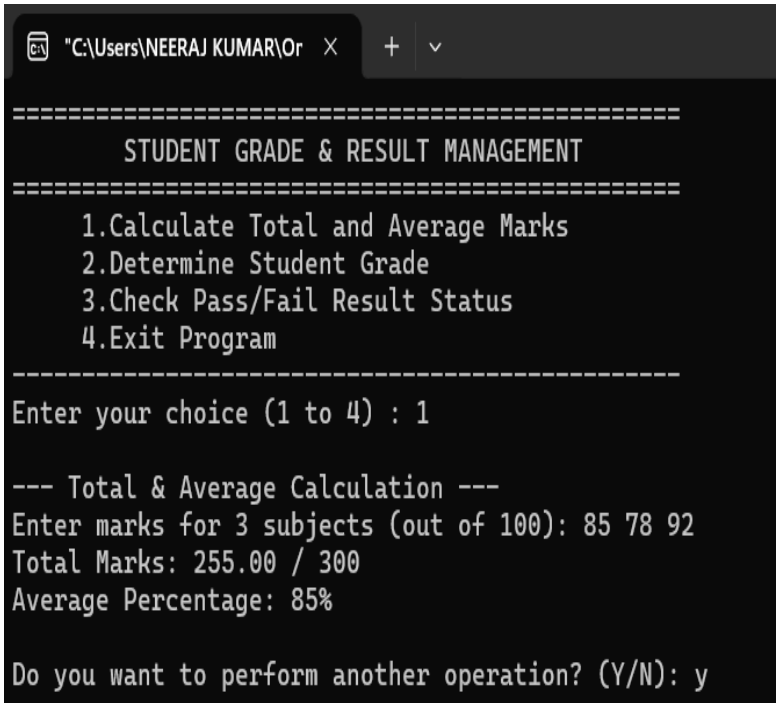
Program:

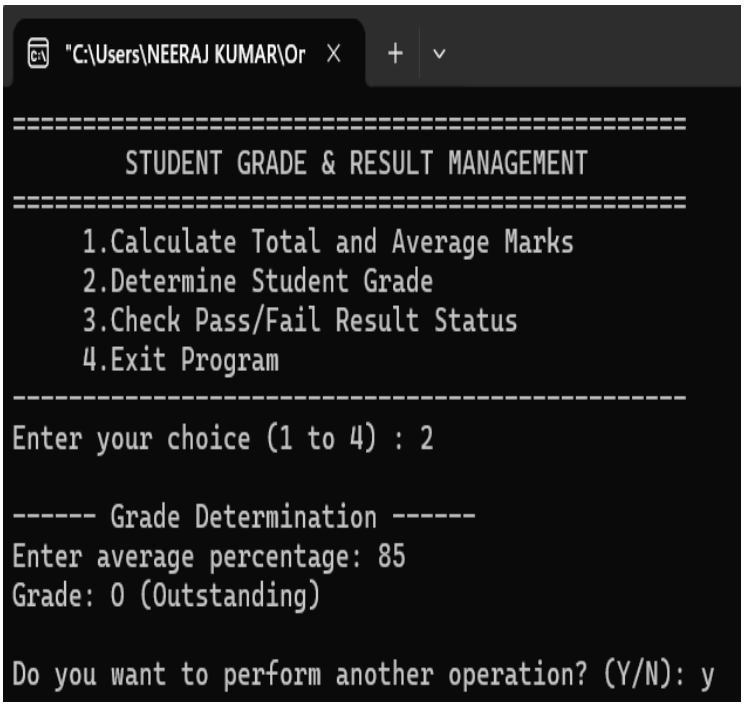
```
#include <stdio.h>
int main()
{
int choice;
float m1, m2, m3, total, avg;
char continue_choice;
// Label using goto to create a menu loop
main_menu:
printf("================================================\n");
printf(" \tSTUDENT GRADE & RESULT MANAGEMENT \n");
printf("================================================\n");
printf("     1.Calculate Total and Average Marks\n");
printf("     2.Determine Student Grade\n");
printf("     3.Check Pass/Fail Result Status\n");
printf("     4.Exit Program\n");
printf("------------------------------------------------\n");
printf("Enter your choice (1 to 4) : ");
if (scanf("%d", &choice) != 1) {
printf("Invalid input! Please enter a valid integer.\n");
return 1;
}
switch (choice) {
case 1:
printf("\n--- Total & Average Calculation ---\n");
printf("Enter marks for 3 subjects (out of 100): ");
scanf("%f %f %f", &m1, &m2, &m3);
if (m1 < 0 || m1 > 100 || m2 < 0 || m2 > 100 || m3 < 0 || m3 > 100) {
printf("Error: Marks must be between 0 and 100.\n");
} else {
total = m1 + m2 + m3;
avg = total / 3.0;
printf("Total Marks: %.2f / 300\n", total);
printf("Average Percentage: %2.f%%\n", avg);
}
break;
case 2:
printf("\n------ Grade Determination ------\n");
printf("Enter average percentage: ");
scanf("%f", &avg);
if (avg < 0 || avg > 100) {
printf("Invalid percentage entered.\n");
} else if (avg >= 85) {
printf("Grade: O (Outstanding)\n");
} else if (avg >= 75) {
printf("Grade: A+ (Excellent)\n");
} else if (avg >= 65) {
printf("Grade: A (Very Good)\n");
} else if (avg >= 55) {
printf("Grade: B+ (Good)\n");
} else if (avg >= 50) {

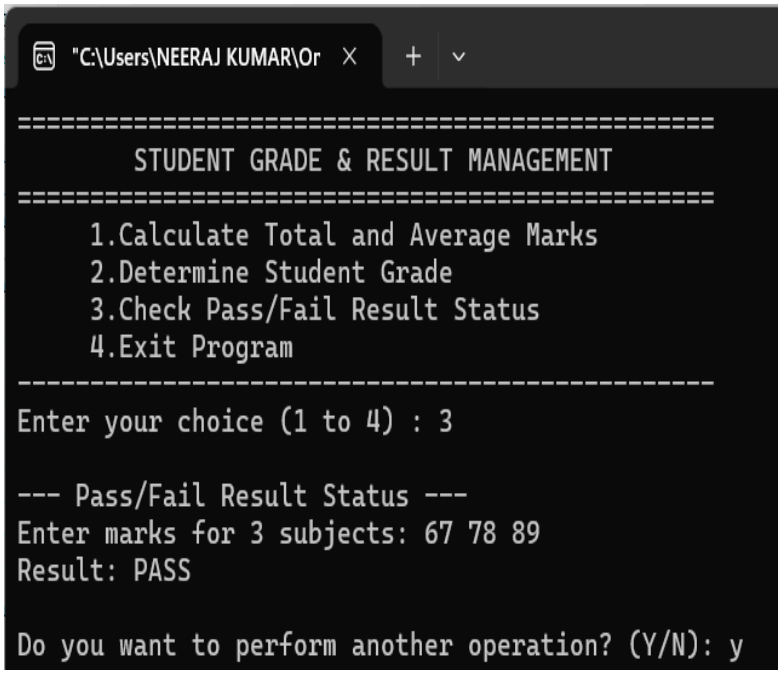
printf("Grade: B (Above Average)\n");
} else if (avg >= 40) {
```

```
printf("Grade: C (Pass)\n");
} else {
printf("Grade: F (Fail)\n");
}
break;
case 3:
printf("\n--- Pass/Fail Result Status ---\n");
printf("Enter marks for 3 subjects: ");
scanf("%f %f %f", &m1, &m2, &m3);
if (m1 < 40 || m2 < 40 || m3 < 40) {
printf("Result: FAIL (Scored below passing mark 40 in one or more
subjects)\n");
} else {
printf("Result: PASS\n");
}
break;
case 4:
printf("\nExiting the system. Thank you!\n");
goto exit_program;
default:
printf("\nInvalid Choice! Please select an option between 1 and 4.\n");
}
printf("\nDo you want to perform another operation? (Y/N): ");
scanf(" %c", &continue_choice);
if (continue_choice == 'y' || continue_choice == 'Y') {
goto main_menu; // Jump back to the main menu using goto
}
exit_program:
return 0;
}
```

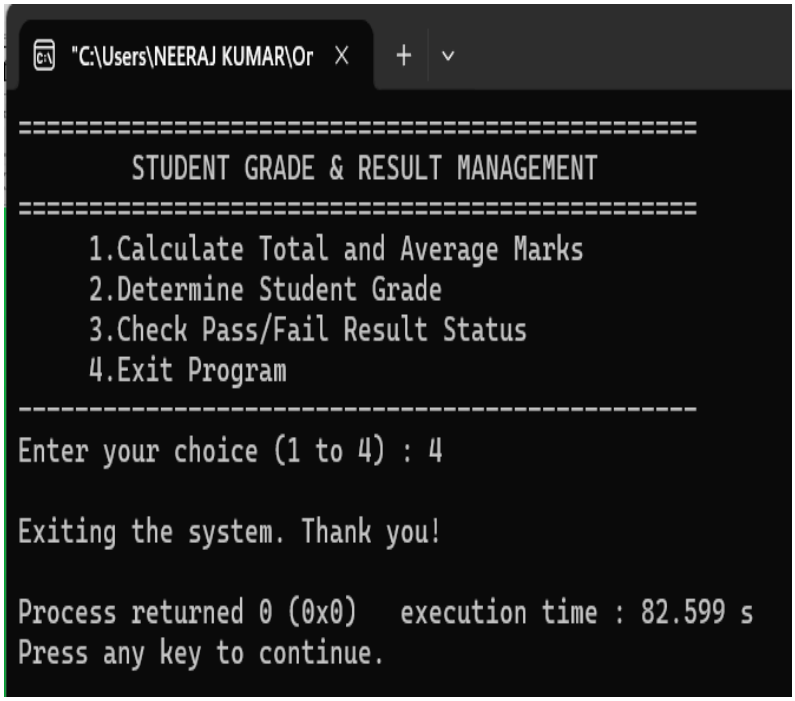
Output:






"C:\Users\NEERAJ KUMAR\Or
==========
STUDENT GRADE & RESULT MANAGEMENT
==========
1.Calculate Total and Average Marks
2.Determine Student Grade
3.Check Pass/Fail Result Status
4.Exit Program

Enter your choice (1 to 4) : 3
--- Pass/Fail Result Status ---
Enter marks for 3 subjects: 67 78 89
Result: PASS
Do you want to perform another operation? (Y/N): y


"C:\Users\NEERAJ KUMAR\Or
==========
STUDENT GRADE & RESULT MANAGEMENT
==========
1.Calculate Total and Average Marks
2.Determine Student Grade
3.Check Pass/Fail Result Status
4.Exit Program

Enter your choice (1 to 4) : 4
Exiting the system. Thank you!
Process returned 0 (0x0) execution time : 82.599 s
Press any key to continue.